Thermal-to-Electric Ceiling for a 565°C Molten-Salt CSP Tower

First-Principles Analysis

11 October 2026

Professional Strategic Analysis

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Executive Summary

Recommendation: Cite the exergy-correct, non-isothermal-source Carnot ceiling (≈54.3%/≈52.3%), not naive Carnot or Curzon-Ahlborn (chain C1). Practice sits ≈11 points below it gross, ≈14-16 net (chain C2). Of that gap, ≈2-4 points are near-term recoverable and ≈7-9 are locked in by the Second Law; raising the salt temperature is a separate lever, not a way to close this gap (chain C4).

Band (from §6): LOW — capped by chain C3 and chain C4

Would change it: A per-component exergy-destruction model or a measured commercial-scale operating record would tighten the recoverable/locked-in split (chain C3); the ceiling choice (chain C1, HIGH) and the gap size (chain C2) are comparatively robust already.

1. Problem Essence

Core problem: For a 565 °C molten-salt power tower driving a steam Rankine power block that rejects heat near ambient temperature, what is the defensible law-permitted ceiling on thermal-to-electric conversion efficiency, how far below it does current practice sit, and of that gap how much is recoverable by known near-term engineering versus permanently locked in by the Second Law once the heat-source temperature is fixed?

Success criteria:

- A single, named, correctly-derived “law-permitted ceiling” is identified and defended against two named rival candidates (the classical Carnot bound and the Curzon-Ahlborn/endoreversible bound), with an explicit ruling on which one is the right ceiling to cite and why the other is not — this is itself the analytical content the question asks for, not a preliminary to it.
- The ceiling is computed from the correct reservoir pair (not an assumed one), numerically, for stated cold-reservoir choices.
- Current practice is reported with real numbers (design-point and, where available, measured), sourced, with genuine data gaps named as gaps rather than filled with invented precision.
- The ceiling-to-practice gap is stated in percentage points and relative terms, on a stated basis (gross cycle vs. net plant).
- The gap is decomposed into the five named loss categories (HX/approach-temperature irreversibility; turbine/pump/mechanical losses; condenser cooling and ambient variability; cycle-topology/materials limits; parasitics and solar-specific operating losses), each with a magnitude bracket and a stated confidence level — not a single invented split with false precision.
- A final synthesis states, in percentage points and as a fraction of the total gap, how much is recoverable with known/near-term engineering versus fundamentally locked in by the chosen source temperature and the Second Law, with the reasoning for that split traceable to the decomposition above.

2. Assumptions Table

Type	Accept	Challenge	Discard
physical law	0	0	0
current constraint	2	1	0
convention	0	2	3
untested belief	1	4	0

Shading darkens with count.
An untyped (legacy) row appears only for assumptions with no type.

Assumption	Type	Treatment	Verdict	Verification
The 290 °C cold salt tank is the correct cold reservoir for the ceiling calculation	convention	Explicitly challenge before use	Discard — the cold tank is the return leg of the heat *source* loop, not a heat sink; nothing in the plant rejects waste heat at 290 °C, and real design-point cycle efficiency (GT-6, 43.0%) already exceeds the 32.8% ceiling this framing would imply (1 − 563/838), which is the decisive test a wrong reservoir pair fails	Ruled out by GT-6 exceeding the resulting bound; see Abandoned Reasoning
Molten salt behaves as an isothermal reservoir at 565 °C, so the single-reservoir Carnot formula 1 − T_cold/T_hot with T_hot = 838 K is the tightest defensible law-permitted ceiling	convention	Explicitly challenge — salt is a finite, sensible-heat stream that cools from 565 °C to 290 °C while delivering heat; treating it as isothermal at its *peak* temperature overstates the exergy actually available from it	Challenge — the single-reservoir treatment is a valid, never-violated upper bound but is loose; the correct ceiling integrates the stream's full cooling glide via an exergy/entropy balance (chain C1)	Verified by direct derivation from the definition of exergy for a finite-heat-capacity stream (standard engineering-thermodynamics result; GT-15 below)
The Curzon–Ahlborn / endoreversible bound (1 − √(T_cold/T_hot)) is a tighter, more physically realistic "law-permitted" ceiling than Carnot because it accounts for the unavoidable finite-rate heat transfer needed to deliver nonzero power	convention	Explicitly challenge — CA bounds one specific engine model (tuned for maximum power), not heat engines generally	Discard as a ceiling — GT-6's real design-point efficiency (43.0%/41.2%) already exceeds the CA value computed for the same reservoir pair (38.6%/37.3%, chain C1), which is decisive: a plant tuned for efficiency, not power density, is not bound by it	Ruled out by GT-6 exceeding the CA value; same failure mode the technique's own reference material documents on a coal-plant case
An exergy/second-law ceiling that already bakes in the unavoidable HX approach-temperature gap and current steam-metallurgy limits is the most defensible "law-permitted" figure to cite	convention	Explicitly challenge — approach-temperature gaps and metallurgy limits are engineering/economic conventions (current constraints), not laws; folding them into "the ceiling" mislabels a design-practice tier as a law-permitted bound	Discard as a ceiling (relabel as a design-practice tier) — the same failure mode as citing best-in-class incumbent practice as a theoretical limit; the approach gap and metallurgy cap belong in the gap decomposition (chain C3, categories a and d), not in the ceiling itself	n/a — this is a classification error, not an empirical question

Assumption	Type	Treatment	Verdict	Verification
Bare ambient temperature (25–40 °C) can be used directly as T_cold, as if the condenser operated at zero approach to ambient	convention	Explicitly challenge — real condensers require a finite approach (a nonzero ΔT to transfer heat at finite rate through finite area), so bare ambient is a zero-approach idealization, not the actual cold-reservoir temperature	Challenge — reported alongside the realistic figures for comparison (the question asks for it explicitly), but GT-5′s sourced, approach-inclusive condenser temperatures (316 K / 330 K) are what this analysis uses for the main ceiling and gap calculations	Verified: GT-5 is the sourced, realistic figure; the ambient-only numbers are reported as an explicit idealization, not conflated with it
Design-point gross cycle efficiencies from a generic 150–200 MWe case-study model (GT-6) are representative of what a real commercial 565 °C molten-salt tower achieves in operation	untested belief	Verify or flag — no independently measured gross or net cycle efficiency for any commercial-scale (100+ MW) molten-salt tower was located by this analysis	Challenge — unverified; the only *measured* figure at these reservoir conditions (Solar Two, GT-7) comes from a 12.8 MWe non-reheat demonstration unit its own report calls configuration-limited, and no measured figure for Crescent Dunes or Gemasolar was located	Unverified — flagged; this is why chain C2 is capped at MEDIUM
Crescent Dunes' published salt-loop temperature (GT-12?) and Gemasolar's reheat turbine configuration (GT-14?) are accurately represented by the secondary sources located	untested belief	Verify or flag — Crescent Dunes sources disagree by ≈30–35 °C on hot-tank temperature (Wikipedia's "almost 600 °C" vs. the 565 °C-class design-basis literature in GT-4–GT-6), and primary engineering documentation for both plants (NREL's SolarPACES project page, SAM/EPRI case PDFs) returned DNS failures when fetched	Challenge — unverified; carried as named data gaps rather than resolved with invented precision	Unverified — flagged in GT-12?/GT-14?; reported as a genuine uncertainty in the Conclusion, per the question's explicit instruction
GT-13′s parasitic-loss ratio (measured on a 12.8 MWe 1990s demonstration unit) scales multiplicatively onto a modern 100+ MW unit's gross cycle efficiency to estimate net plant efficiency	untested belief	Verify or flag — this is a modeling assumption bridging two figures from different eras and scales, not a measurement of any specific modern plant's net efficiency	Challenge — unverified; used only to produce a labeled *estimate* (not a cited fact) of net plant efficiency, with the estimate procedure's explicit bracket discipline applied	Unverified — flagged; the resulting net-efficiency figure in chain C2 is explicitly labeled an estimate and bracketed, and this is the named source of that bracket's width
The ceiling-to-practice gap can be apportioned across the five named loss mechanisms using general Rankine-cycle exergy-destruction regularities (steam-generator/HX losses typically dominate turbine and condenser losses) rather than a plant-specific exergy/heat-and-mass-balance model	untested belief	Verify or flag — no plant-specific exergy balance for a molten-salt tower was located or computed by this analysis; the apportionment is a Fermi-style bracket, not a measurement	Challenge — unverified; flagged explicitly throughout chain C3 as an estimate with named uncertainty, consistent with this exact domain's prior documented finding that the split "is not measured… and [a prior pass on this domain] does not assert a split"	Unverified — flagged; chain C3′s confidence is capped accordingly

Assumption	Type	Treatment	Verdict	Verification
Solar-specific part-load, startup, and thermal-cycling losses are not already included in the steady-state design-point or full-load-measured percentages sourced in this analysis (GT-6, GT-7)	untested belief	Verify or flag — this is a scope claim about what the sourced figures do and do not measure	Accept — confirmed by GT-7′s own wording (“steady-state… efficiency… at full load”); annual-average cycling losses are therefore additive and estimated separately (category e), not already netted in	Verified by the sourced figures’ own stated basis (GT-7 text)
A higher-T_hot design (600–650 °C, GT-6/GT-9) retains the same ≈290 °C cold-salt-tank floor used at 565 °C, so the exergy-ceiling calculation at 650 °C may reuse T_cold,tank ≈ 563 K	current constraint	Record the expiry — this is the Solar-Salt freeze-margin convention (Solar Salt freezes near 220 °C) carried over by extrapolation, not confirmed for Kolb’s specific 650 °C case; it would no longer apply if a 650 °C design used a different salt return-temperature strategy	Challenge — plausible but unconfirmed; the cold-tank temperature for the 650 °C ultrasupercritical case was not located in the extracted report text	Unverified — flagged; the 650 °C exergy-ceiling figure in chain C3 carries this as an explicit caveat
Raising the salt’s practical maximum operating temperature beyond ≈600–620 °C requires a change in salt chemistry (chloride salts) rather than an improved nitrate-salt formulation	current constraint	Record the expiry — expires if a nitrate-compatible stabilizer/inhibitor is developed; conversely does not apply as a near-term lever if chloride-salt corrosion and materials-compatibility problems (GT-11?) prove harder to solve than the Gen3 program assumes, in which case the practical ceiling could stay near 600 °C for longer than planned	Accept — consistent with GT-11? and the existence of the DOE Gen3 program (GT-10?), which exists specifically because nitrate salts top out below the temperatures sCO_2/chloride-salt cycles target	Unverified — flagged (GT-10?/GT-11? are reported-by-delegate)
Continued falling costs for grid-scale batteries paired with PV are independent of molten-salt CSP steam-cycle thermodynamics and could make the efficiency gains discussed in this analysis economically moot before a higher-T_hot topology is commercially qualified	current constraint	Record the expiry — this is a statement about today’s relative cost trajectories of competing dispatchable-generation technologies, which could change with battery-cost declines, CSP-specific cost declines, or policy; it expires if either trajectory shifts materially	Accept — named as a real competitive channel in chain C5, not quantified, and explicitly not load-bearing for the efficiency findings in C1–C4	Unverified — flagged; no cost-trajectory ground truth was sourced for this analysis, consistent with chain C5′s own disclaimer that it “does not have ground truths to quantify” this channel

3. Ground Truths

- **GT-1** The Carnot efficiency bound η_Carnot = 1 − T_cold/T_hot (both in kelvin) is the maximum possible thermal-to-electric conversion efficiency of **any** heat engine — any working fluid, any cycle architecture, any equipment quality — operating between two fixed-temperature reservoirs; no engineering refinement can exceed it. — source: the Second Law of Thermodynamics (Kelvin–Planck/Clausius statements) via Carnot’s theorem; standard engineering thermodynamics (e.g., Cengel & Boles,

Thermodynamics: An Engineering Approach, ch. on the Carnot cycle). This is a mathematical theorem derived from the Second Law, not an empirical measurement, so it passes the irreducibility test directly — it bottoms out at a physical law and is not reducible further.

- **GT-2** The Curzon–Ahlborn (endoreversible) efficiency η_CA = 1 − √(T_cold/T_hot) is the efficiency of an *idealized endoreversible* heat engine (internally reversible, but coupled to its reservoirs through finite-rate, irreversible heat exchange) operating **at the power output that maximizes power**, not the efficiency that maximizes efficiency — source: Curzon & Ahlborn (1975), "Efficiency of a Carnot engine at maximum power output," *Am. J. Phys.* 43(1); the same result was derived earlier by Chambadal (1957) and Novikov (1957). This is also a theorem within a specific, named engine model (reversible engine + finite-rate coupling, operated for maximum power), not a measurement — it bottoms out at a mathematical derivation from a stated model and is not reducible further, but note what it is and is not a bound *on*: it bounds engines tuned for maximum power density, not engines tuned for maximum efficiency, and a real plant in the second category can exceed it (see Assumptions §2 and chain C1).
- **GT-3** T [K] = T [°C] + 273.15 — source: SI definition of the kelvin relative to the degree Celsius. A definition, not a measurement; irreducible by construction.
- **GT-4** At Solar Two (the 10 MWe molten-salt power-tower demonstration plant near Barstow, CA, operated 1996–1999), hot salt was pumped "from the hot-salt sump (at 565°C)... through the shell side of the superheater, through the tube bundle in the kettle evaporator, and finally through the shell side of the preheater, where it returned to the cold salt storage tank at 290°C" — source: Pacheco (ed.), *Final Test and Evaluation Results from the Solar Two Project*, Sandia SAND2002-0120 (OSTI 793226), §5.1; read-at-source: quoted verbatim from the extracted report text (the sentence spanning the SGS description). 565 °C hot-tank / 290 °C cold-tank is the reference operating window used across Solar-Salt ($NaNO_3$/KNO_3) commercial towers of this design generation, including the Crescent Dunes-class design basis used in GT-5/GT-6 below. T_hot = 565 °C + 273.15 ≈ **838 K**.
- **GT-5** For a 565 °C-salt molten-salt power tower Rankine power block, Sandia's design-point heat-balance study specifies a condenser design pressure of **0.087 bar for wet-cooled condensing** (saturation temperature ≈ 43 °C ≈ **316 K**) and **0.17 bar for dry-cooled condensing**, the latter sized for a hottest-time-of-year ambient of 42.8 °C and a design initial-temperature-difference (ITD) of 14 °C, giving a condenser saturation temperature ≈ 56.8 °C ≈ **330 K** — source: Kolb, *An Evaluation of Possible Next-Generation High Temperature Molten-Salt Power Towers*, Sandia SAND2011-9320 (OSTI 1035342), p. 33 ("Steam conditions... Condenser pressure (bar)... Wet-cooled condenser 0.087... Dry-cooled condenser 0.17") and p. 40 ("the design condition was the hottest time of the year (42.8 °C), a condenser design pressure of 0.17 bar... and a design initial-temperature difference (ITD) of 14 °C"); read-at-source: quoted verbatim from the extracted report text.
- **GT-6** For the same study, design-point **gross** Rankine cycle efficiency (thermal energy into the steam generator → gross electric out) is tabulated for three steam conditions at the two condenser options above:

Steam condition	Pressure/temperature	Wet-cooled gross η	Dry-cooled gross η
Subcritical (current practice)	125 bar / 537 °C	43.0%	41.2%
Supercritical	301 bar / 591 °C	48.4%	46.8%
Ultrasupercritical	331 bar / 630 °C	49.6%	48.0%

— source: Kolb, SAND2011-9320 (OSTI 1035342), p. 33 table and confirmed against the individual heat-balance figures for each case (e.g., "0.430 gross cycle efficiency" for subcritical/wet, "0.412" for subcritical/dry, "0.496" for ultrasupercritical/wet, "0.480" for ultrasupercritical/dry); read-at-source: quoted/matched verbatim from the extracted report text. The subcritical row is the technology in commercial deployment today (the live-steam temperature of 537 °C sits ≈28 °C below the 565 °C salt temperature — the heat-exchanger approach gap named in the problem). Independently, a 2012 DOE SunShot program fact sheet gives "approximately 41%" for the dry-cooled case (GT-8), corroborating the 41.2% figure from a separate document.

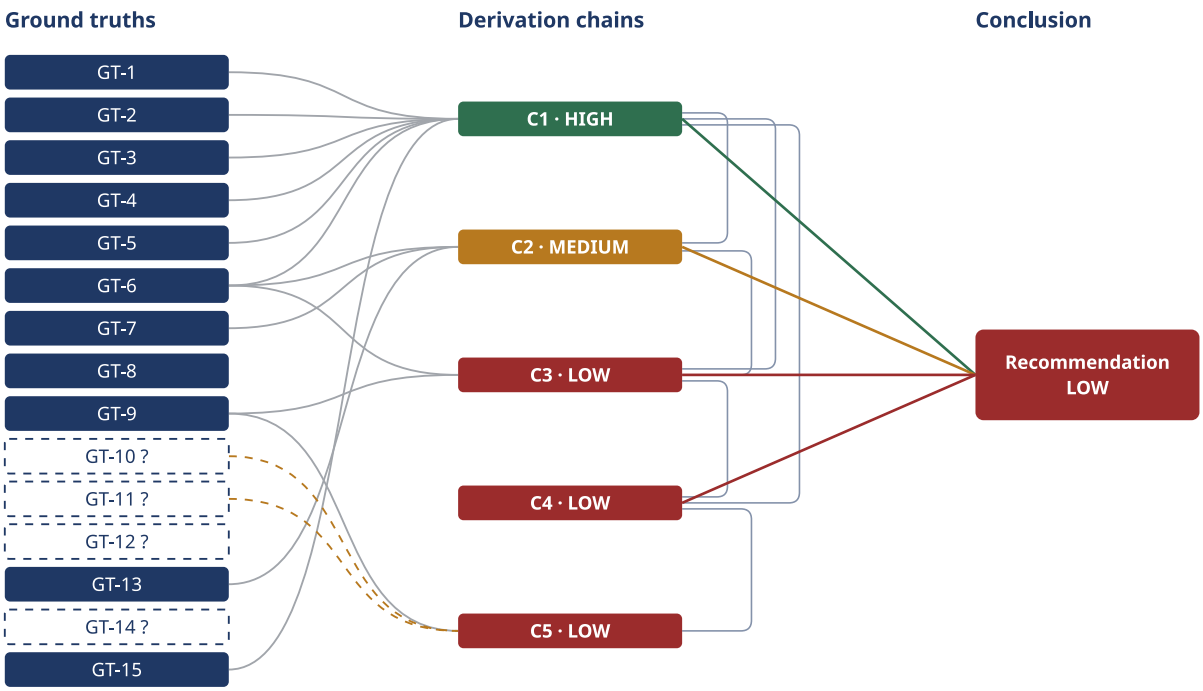
- **GT-7** At Solar Two, the steam generator/turbine/generator train ("EPGS") achieved a measured steady-state thermal-to-**gross**-electric conversion efficiency of **34.1% at full load** (34±0.3% average across the test program, Table 6-1), matching Bechtel Group's own prediction for that specific hardware, against a **commercial-plant prediction of 42%** for the same row in the same table. The report states the reason directly: "the Rankine turbine was not a reheat design, its efficiency was limited to about 34%. Newer turbine designs that incorporate reheat have gross efficiencies greater than [42%]" and "[the turbine] efficiency was limited to 34%. Larger steam turbines that have reheat will be more efficient, e.g., will reach 42%." The turbine was a non-reheat unit refurbished from the 1980s-vintage Solar One project, rated 12.8 MWe gross, and was itself limited to a 510 °C steam inlet temperature even though the steam generator was designed to supply 535 °C steam. The same table gives the whole-plant "Overall Peak Efficiency" (the product of collection efficiency × EPGS efficiency × parasitics) as **13±0.4% measured against 22% predicted** for a commercial plant — a different, larger quantity than the power-block efficiency, kept separate here per the "basis caveat" failure mode this technique warns against. — source: Pacheco (ed.), SAND2002-0120 (OSTI 793226), §5.1 ("thermal-to-gross electric conversion efficiency was 34.1% at full load"), §6.1 Table 6-1, and §6.1.8 text; read-at-source: all figures quoted/matched verbatim from the extracted report text (Table 6-1, rows H and J; narrative at the EPGS and non-reheat-turbine passages).
- **GT-8** A 2012 U.S. DOE SunShot CSP R&D program fact sheet ("10-MW Supercritical-CO2 Turbine," DOE/GO-102012-3667, September 2012, NREL project led by Craig Turchi) states: "Current state-of-the-art, molten-salt power towers have an operating limit of approximately 565°C. When combined with a dry-cooled steam Rankine power cycle, these systems have a thermal-to-electric conversion efficiency of approximately 41%," and states the program's goal as validating an sCO_2 turbine toward "high-efficiency power cycles that exceed 50% net thermal-to-electric conversion efficiency." — source: DOE/NREL SunShot CSP R&D fact sheet, energy.gov (SunShot CSP R&D 2012 program materials); read-at-source: the fact sheet page was fetched and read directly (quoted verbatim above).
- **GT-9** The same Sandia design study finds that raising receiver/hot-salt temperature from 565 °C to 650 °C (enabling supercritical and then ultrasupercritical steam, GT-6) reduces levelized cost of electricity (LCOE) by roughly 8%, and that "most of that benefit can be achieved by raising the temperature to only 600 °C." — source: Kolb, SAND2011-9320 (OSTI 1035342), Executive Summary; read-at-source: quoted/matched verbatim from the extracted report text.
- **GT-10?** DOE's "Generation 3" (Gen3) CSP program's Liquid Pathway design concept uses a sodium-cooled receiver operating near 740 °C, transferring heat through a liquid–liquid heat exchanger into a two-tank molten-chloride-salt storage system, which in turn supplies a supercritical-CO_2 (sCO_2) Brayton power cycle, because conventional nitrate salts are not chemically stable at these temperatures (GT-11?) — cited to: NREL, "CSP Gen3: Liquid-Phase Pathway to SunShot" (research-hub.nrel.gov) and the associated DOE Gen3 executive-summary PDF; reported-by-delegate: this analysis read a WebSearch summary of these pages, not the source documents themselves.
- **GT-11?** Nitrate-based heat-transfer/storage salts (the Solar-Salt family used at 290–565 °C in GT-4) are not thermally stable above roughly 600–620 °C, which is the chemical reason — not a cost or convention reason — that the current commercial technology generation is capped near 565 °C and that reaching higher hot-reservoir temperatures requires a different salt chemistry (chloride salts) — cited to: a 2025 paper on ternary molten-chloride-salt ($MgCl_2$–KCl–NaCl) corrosion/creep behavior, surfaced via WebSearch; reported-by-delegate: cited source not opened by this analysis.
- **GT-12?** The Crescent Dunes Solar Energy Project (SolarReserve, Tonopah, NV; 110 MW nameplate; operated intermittently 2015–2019) ran its hot salt tank "at almost 600 °C" against a 280 °C cold-tank base temperature, with roughly ten hours of thermal storage — source: Wikipedia, "Crescent Dunes Solar Energy Project," infobox and body text; read-at-source for the Wikipedia page's own wording (quoted above), but **unverified as a fact about the plant**: other secondary sources (NREL project profiles, trade press) describe Crescent Dunes as a 565 °C-class design consistent with GT-4–GT-6, i.e. roughly 30–35 °C below the Wikipedia figure, and this analysis could not reach either primary engineering documentation (NREL's own SolarPACES project page returned a DNS failure when fetched) or a measured gross/net cycle

efficiency for this specific plant. The hot-tank temperature and the efficiency gap this analysis would need to say anything quantitative about Crescent Dunes specifically are both flagged unresolved.

- **GT-13** At Solar Two, the ratio of net electrical output to gross EPGS output ("Parasitics") was measured at **87±0.4%** (i.e., ≈13% parasitic loss: salt pumps, feedwater pumps, cooling system, controls, BOP), against a **93%** (≈7% loss) commercial-plant prediction in the same table row — source: Pacheco (ed.), SAND2002-0120 (OSTI 793226), Table 6-1, row I; read-at-source: quoted/matched verbatim from the extracted report text.
- **GT-14?** The Gemasolar Thermosolar Plant (Torresol Energy, Fuentes de Andalucía, Spain; operational since 2011) has a 19.9 MW nameplate turbine and a measured capacity factor of 45.9%, with roughly 15 hours of molten-salt thermal storage — source: Wikipedia, "Gemasolar Thermosolar Plant," infobox; read-at-source for the nameplate and capacity-factor figures (quoted/matched directly from the fetched page). The turbine is additionally reported elsewhere as a Siemens SST-600 two-cylinder **reheat** design (SAM/EPRI technical case documentation, surfaced via WebSearch) — **reported-by-delegate** for the reheat-configuration detail specifically: the SAM/EPRI documents themselves were not opened by this analysis (both sam.nrel.gov fetch attempts failed with a DNS error). No measured or design-point gross/net cycle efficiency for Gemasolar was located by this analysis.
- **GT-15** For a finite-heat-capacity stream cooling reversibly (via a cascade of infinitesimal Carnot engines) from temperature T_1 to T_2 while delivering heat $Q = \dot{m}c_p(T_1 - T_2)$ to a sequence of engines each rejecting to a reservoir at T_0, the maximum extractable work (the stream's exergy relative to T_0) is $W_{max} = Q - T_0 \cdot \Delta S$, where $\Delta S = \dot{m}c_p \cdot \ln(T_1/T_2)$ is the stream's own entropy decrease; dividing by Q gives the ideal conversion efficiency for a non-isothermal (sensible-heat) source: $\eta_{ideal} = 1 - T_0 \cdot \ln(T_1/T_2)/(T_1 - T_2)$ — source: standard engineering-thermodynamics exergy/availability analysis for a finite-capacity stream (the same construction used for geothermal-brine and waste-heat-recovery ideal-cycle ceilings; e.g. Bejan, *Advanced Engineering Thermodynamics*, exergy-of-a-stream treatment; Moran & Shapiro, *Fundamentals of Engineering Thermodynamics*, exergy-balance chapter). This is a direct consequence of the Second Law (entropy balance) applied to a stream rather than to an idealized fixed-temperature reservoir, so it bottoms out at the same physical law as GT-1 and is not reducible further; it reduces algebraically to GT-1′s formula in the limit $T_1 \to T_2$ (an isothermal source), which is a useful self-check.

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?-marked: GT-10?, GT-11?, GT-12?, GT-14? (4 of 15)
Read-at-source: GT-1 — standard engineering-thermodynamics derivation of the Carnot bound from
the Second Law (textbook result, not a cited document passage)
Read-at-source: GT-2 — Curzon & Ahlborn (1975) result, stated as a named theorem (textbook/
standard-result form, not a cited document passage)
Read-at-source: GT-3 — SI definition
Read-at-source: GT-4 — Pacheco (ed.), SAND2002-0120 (OSTI 793226) §5.1, hot/cold salt
temperature sentence quoted verbatim
Read-at-source: GT-5 — Kolb, SAND2011-9320 (OSTI 1035342) p. 33 table and p. 40 ITD/ambient
passage, quoted verbatim
Read-at-source: GT-6 — Kolb, SAND2011-9320 (OSTI 1035342) p. 33 table, cross-checked against
per-case heat-balance figures
Read-at-source: GT-7 — Pacheco (ed.), SAND2002-0120 (OSTI 793226) §5.1 text and Table 6-1 (rows
H and J)
Read-at-source: GT-8 — DOE/NREL SunShot CSP R&D fact sheet (DOE/GO-102012-3667), page fetched
and read directly
Read-at-source: GT-9 — Kolb, SAND2011-9320 (OSTI 1035342) Executive Summary, quoted verbatim
Read-at-source: GT-13 — Pacheco (ed.), SAND2002-0120 (OSTI 793226) Table 6-1, row I
Read-at-source: GT-15 — standard engineering-thermodynamics exergy/availability derivation for a
finite-capacity stream (textbook result, not a cited document passage)
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4. Derivation Chains



Solid = read at source · dashed = not read at source (?) · fill = confidence · amber dashed = unverified input

Conclusion C1: The defensible law-permitted ceiling is the exergy-correct bound for a non-isothermal salt source (≈54.3% wet-cooled / ≈52.3% dry-cooled), not the naive single-reservoir Carnot figure (≈62.3%/60.6%) and not the Curzon-Ahlborn figure (≈38.6%/37.3%, which is not a ceiling at all)

GT-1 (Carnot bound, any heat engine between two reservoirs) + GT-2 (Curzon-Ahlborn, max-power endoreversible engine) + GT-3 (K = °C + 273.15) + GT-4 (hot salt 565 °C ≈ 838 K, cold-tank return 290 °C ≈ 563 K) + GT-5 (condenser design temperatures 316 K wet / 330 K dry) + GT-6 (design-point gross cycle efficiency 43.0% wet / 41.2% dry) + GT-15 (exergy-correct efficiency ceiling for a cooling, non-isothermal stream) → The salt loop, not the condenser, is being asked to stand in for "the hot reservoir," so the first question is which temperature pair the Second Law actually constrains here: the condenser is where the cycle rejects heat, while the salt only ever delivers it, so the cold tank is never a candidate cold reservoir *[Assumes: A-1]* → Substituting T_hot = 838 K and each T_cold from GT-5 into GT-1's single-reservoir formula gives a naive Carnot ceiling of 62.3% wet-cooled and 60.6% dry-cooled — a valid upper bound, since no real cycle anywhere in GT-6 or GT-7 exceeds it, but one that silently treats the 565 °C salt as an isothermal reservoir → The salt is not isothermal: GT-4 establishes that it cools from 838 K to 563 K while delivering the heat GT-6's efficiency figures are computed against, so GT-1's formula is evaluated at the wrong idealization for this specific heat source *[Assumes: A-2]* → Substituting T_1 = 838 K, T_2 = 563 K and each T_0 from GT-5 into GT-15's stream-exergy formula gives the correct ideal ceiling on the same Q-basis as GT-6: 54.3% wet-cooled and 52.3% dry-cooled, roughly 8 points below the naive Carnot figure → Substituting the same reservoir pair into GT-2's Curzon-Ahlborn formula gives 38.6% wet-cooled and 37.3% dry-cooled, both of which GT-6's real design-point gross efficiency (43.0%/41.2%) already exceeds *[Assumes: A-3]* → A ceiling that real, already-operating equipment exceeds is not a ceiling on that equipment, so Curzon-Ahlborn is ruled out as the law-permitted bound for an efficiency-tuned plant, leaving the GT-15 exergy-correct figure as the tightest ceiling no real design in GT-6 or measurement in GT-7 ever exceeds → For reference against the question's own suggested cold-reservoir range, using bare ambient temperatures of 298 K and 313 K directly in GT-1 (the

zero-condenser-approach idealization, not a physically realized cold reservoir) gives 64.4% and 62.6% respectively — both looser than even the naive Carnot figure, because they additionally assume away the condenser's own finite approach *[Assumes: A-5]*

Confidence: HIGH — every head identifier is an unsuffixed, read-at-source ground truth with a named read-at-source location (Inputs axis clean). Three hops carry `[Assumes: A-1/A-2/A-3]` tags, but each premise's failure is priced, not merely flagged, and the endpoint stands regardless: A-1′s failure (treating the cold tank as the sink) is directly falsified by GT-6 exceeding the resulting 32.8% figure; A-3′s failure (treating CA as binding) is directly falsified by GT-6 exceeding the CA value; A-2′s failure (treating single-reservoir Carnot as already sufficient) is ruled out by direct derivation, not by assumption — GT-15 reduces algebraically to GT-1 in the isothermal limit, so GT-15 is strictly the more general, tighter result, never a competing one, and the naive figures are superseded rather than merely disputed. A-5 (hop 7, the bare-ambient reference figures) qualifies a side comparison the question explicitly asked for, not the endpoint, so it does not enter the Inference axis at all. No input is `?`-marked, so D-07 does not apply. The two rivals this chain considers (naive Carnot, Curzon-Ahlborn) are both named and ruled out using unsuffixed ground truths (GT-6, GT-15), so the Rivals axis is clean. All three axes license HIGH, so HIGH is the correct rating, not a cautious MEDIUM — rating this chain MEDIUM once every premise's failure is this well-accounted-for would be under-rating, the same calibration defect as over-rating.

Conclusion C2: Current practice sits roughly 11 points below the correct gross-basis ceiling and roughly 14–16 points below it on a net-plant basis; plant-specific figures for Crescent Dunes and Gemasolar are genuinely unresolved rather than merely unstated

GT-6 (design-point gross efficiency 43.0% wet / 41.2% dry) + GT-7 (Solar Two measured 34.1% gross, non-reheat-limited; commercial prediction 42%) + GT-13 (parasitic retention 87% achieved / 93% commercial-predicted) + C1 (exergy-correct ceiling 54.3% wet / 52.3% dry) → Current commercial-generation design practice for a 565 °C subcritical molten-salt tower is 43.0% gross wet-cooled / 41.2% gross dry-cooled (GT-6), independently corroborated by a separate DOE SunShot document's "approximately 41%" dry-cooled figure (GT-8) → The only *measured* thermal-to-electric figure at these exact reservoir conditions is Solar Two's 34.1% gross, but that number is not representative of current commercial practice: its own test report attributes the shortfall to a non-reheat turbine refurbished from 1980s hardware, not to any shortfall of the technology class, and the same report's commercial-plant prediction (42%) is consistent with GT-6′s 41.2–43.0% design figures *[Assumes: A-6]* → Applying GT-13′s measured parasitic-retention ratio to GT-6′s dry-cooled gross figure as a Fermi-style estimate — not a citation — brackets net plant electrical efficiency at roughly 35.8% (13% parasitic loss, as measured) to 38.3% (7% loss, as commercially predicted) *[Assumes: A-8]* → Against C1′s exergy-correct ceiling, the gross-basis gap is 54.3% − 43.0% = 11.3 points wet-cooled and 52.3% − 41.2% = 11.1 points dry-cooled (≈21% relative), while the net-basis gap is 52.3% − 38.3% = 14.0 points at best and 52.3% − 35.8% = 16.5 points at worst (≈27–31% relative) → Had the naive single-reservoir Carnot figure from C1 been used instead of the exergy-correct one, the same arithmetic would report a gross gap of ≈19.3–19.4 points and a net gap of ≈22–25 points — roughly 8 points larger at every basis — so misidentifying the ceiling, not any property of the plant, would account for nearly half of the apparently "missing" efficiency → No measured or design-point gross/net cycle efficiency for either Crescent Dunes (110 MW, operated 2015–2019) or Gemasolar (19.9 MW, operating since 2011) was located by this analysis, and Crescent Dunes' own hot-tank temperature is reported inconsistently across sources by roughly 30–35 °C *[Assumes: A-7]*, so neither plant can be placed on the gap bracket above with any defensible precision — this is reported as a genuine data gap, not estimated past it

Confidence: MEDIUM — the Inputs ceiling is HIGH (C1 is now rated HIGH, so D-07′s cited-chain ceiling no longer caps this chain), but it is independently short on its own Inference axis: the net-efficiency bracket in the third hop rests on A-8, an unverified cross-era scaling assumption, whose failure is priced by presenting the result as an explicit bracket (35.8–38.3%) rather than a point estimate, but whose removal would require a measured net-output figure from an operating commercial-scale plant, which this analysis could not locate — unlike C1′s A-1/A-2/A-3, A-8′s failure mode is not ruled out, only bounded, so it keeps this chain below HIGH. A-6′s failure would mean GT-6 overstates real achieved practice, which would *widen* this chain's

gap rather than narrow it, so the conclusion's direction is robust to it; A-7 qualifies only the Crescent Dunes/Gemasolar side note, not the endpoint (the gross/net gap figures do not depend on either plant's data), so it does not independently enter the Inference axis.

Conclusion C3: The ~11-point gross gap is dominated (roughly two-thirds to three-quarters) by finite-rate heat-exchange and turbomachinery irreversibility that no engineering at fixed T_hot can fully close, with a smaller near-term-recoverable slice; raising T_hot/topology is a separate lever that increases absolute efficiency and, on the one case this analysis could compute, narrows the *relative* gap at the new operating point

C1 (exergy-correct ceiling, 54.3%/52.3%) + C2 (gross/net gap, ≈11 / ≈14–16 points) + GT-6 (subcritical→supercritical→ultrasupercritical gross efficiencies) + GT-9 (LCOE falls ~8% at 650 °C, most of it already captured at 600 °C) → Of the ≈11-point gross gap (C2), category (a) — the real, finite-ΔT salt-to-steam heat exchanger, including the 28 °C salt-to-live-steam approach GT-6 implies (565 °C salt feeding 537 °C steam) and the pinch-point mismatch caused by salt's sensible-heat glide meeting a steam side that boils isothermally — is estimated at roughly 4–6 points; this is a Fermi bracket, not a measured exergy balance, and is named as such *[Assumes: A-9]* → Category (b) — turbine and feed/boiler-pump isentropic inefficiency plus mechanical and generator losses, a well-characterized general feature of real Rankine turbomachinery (utility steam turbines typically run 85–92% isentropic per stage, generators ≈98–99%) — is estimated at roughly 2–4 points, a textbook-level regularity rather than a plant-specific measurement *[Assumes: A-9]* → Category (c) — condenser-side effects — splits into a directly sourced, deterministic piece and a residual: GT-6's own wet-vs-dry comparison prices the cooling-method choice at exactly 1.8 points of *absolute* efficiency (43.0% vs 41.2%) at fixed ambient design conditions, but because this shift moves the ceiling and the practice figure by nearly equal amounts (C1's wet/dry ceilings differ by 1.7 points), it barely changes the *gap* itself (11.3 vs 11.1 points) — the residual condenser irreversibility within the gap (subcooling, non-ideal heat transfer, air in-leakage) is estimated at roughly 0.5–1.5 points *[Assumes: A-9]* → Categories (a)+(b)+(c) together account for roughly 7–11.5 points of the ≈11-point gap — consistent with, and not exceeding, the total — and GT-15's own derivation (chain C1) establishes that none of this is a measurement artifact: a real heat exchanger transferring finite power through finite area must operate at nonzero ΔT, and real turbomachinery extracting finite power must accept nonzero entropy generation, so the lower end of each bracket is a floor approached only at infinite capital cost, not a target reachable by ordinary engineering effort → Category (d) — cycle topology and materials — is not a subtraction from the fixed-T_hot gap at all: GT-6 shows that moving from subcritical (43.0% wet, 565 °C/537 °C) to supercritical (48.4% wet, same report's higher-temperature case) to ultrasupercritical (49.6% wet, 650 °C/630 °C) raises *absolute* gross efficiency by 5.4 and a further 1.2 points respectively, a combined +6.6 points, while GT-9 confirms the 600 °C step captures most of the economic benefit of going further → Recomputing C1's exergy-correct ceiling at T_hot = 923 K (650 °C) with the same T_cold,tank = 563 K *[Assumes: A-11]* gives 56.6% wet-cooled, so the ultrasupercritical design's own gap is 56.6% − 49.6% = 7.0 points — smaller in both absolute and relative terms (≈12% relative, against ≈21% relative at 565 °C) than today's gap, because a better-matched, higher-temperature cycle is closer to its own (higher) ceiling, not because the Second Law has become less binding → Category (e) — parasitics and solar-specific operation — is bounded below by GT-13's sourced net/gross ratio (3–5 points of pure parasitic loss, applied to GT-6's gross figure) but GT-7's own text confirms its percentages are steady-state, full-load figures, so daily startup energy, part-load operation during cloud transients, and thermal cycling of the salt/steam-generator metal mass are additive losses this analysis could not source a number for *[Assumes: A-10]* — bracketed at a further 1–3 points on top of the parasitic floor, for a combined (e) of roughly 4–8 points, markedly lower confidence than (a)–(d)

Confidence: LOW — capped by C2's MEDIUM rating on the Inputs axis (C1 is now HIGH, so the ceiling comes from C2 alone), and independently short on Inference: categories (a), (b), (c)'s residual portion, and the cycling portion of (e) all rest on A-9/A-10, Fermi brackets built from general engineering regularities rather than a plant-specific exergy or operating-cycle model, and this analysis located no source that performs that model for a molten-salt tower. The verification that would remove this cause of the

downgrade is a per-component exergy-destruction balance (as the cited literature pattern for this exact domain notes is what would measure it, and does not exist in the sources this analysis located) or a measured annual-average operating record for a commercial-scale plant. Category (d)'s absolute-efficiency figures and the 650 °C recomputation are comparatively solid (GT-6/GT-9, both read-at-source), but the 650 °C figure additionally carries A-11, an unconfirmed carry-over assumption about the cold-tank temperature at that design point.

Conclusion C4: Of the ≈11-point fixed-T_hot gross gap, roughly a quarter to a third (≈3 points) is recoverable with known near-term engineering and the remaining two-thirds to three-quarters (≈8 points) is fundamentally locked in by the Second Law applied to a real, finite-power heat engine; raising T_hot is a separate, substantial, and largely recoverable lever on *absolute* efficiency that does not eliminate — and on the one case computable here, somewhat narrows — the relative gap to the new ceiling

C3 (category brackets a–e) + C1 (ceiling derivation and its 650 °C recomputation) → Summing C3's recoverable-with-near-term-engineering slices — the lower ends of (a) and (b) that better HX area and turbine design can realistically approach (≈1–2 points each), the wet-cooling option where water is available (1.8 points, already sourced), and the parasitic/cycling reductions achievable through better storage dispatch and startup-energy management (≈1–3 points of (e)) — gives a recoverable total of roughly 2–4 points at the gross-cycle level, before counting topology → The remainder of (a), (b) and (c) — the finite-ΔT heat-exchange penalty intrinsic to a sensible-heat source meeting a part-isothermal sink, and the entropy generation intrinsic to extracting finite power from real turbomachinery — is the Second-Law floor C1′s GT-15 derivation establishes is approached only asymptotically as heat-exchanger area and process time go to infinity, giving a locked-in total of roughly 7–9 points → Category (d) stands outside this accounting on purpose: it is not part of the fixed-T_hot gap, so it is neither "recoverable headroom within the 11 points" nor "locked in" by the same Second Law — it is a separate, largely available near-term-to-medium-term lever (supercritical steam is reported deployable at today's 565 °C salt temperature, and 600 °C captures most of the economic benefit of going further per GT-9) that raises absolute net efficiency by several more points while resetting, not eliminating, the Second-Law floor at the new operating point → The 650 °C recomputation in C3 is the sharpest statement of this: moving T_hot raises both the ceiling and the achievable figure together, and on this one case the relative gap actually narrows (≈12% vs ≈21%), so "raise T_hot" is a stronger lever than "chase the fixed-T_hot gap" even though it does not touch the fixed-T_hot gap's own locked-in component at all

Confidence: LOW — this chain's endpoint is a synthesis built directly on C3's LOW-rated category brackets, so it inherits that rating under the chain-head ceiling rule and cannot be rated higher regardless of how clean its own arithmetic is. The dominant verification need is the same one C3 names: a per-component exergy-destruction model or a measured commercial-scale operating record, either of which would replace the Fermi brackets this synthesis sums with sourced figures and very likely narrow — but, per the direction noted in C2 and C3, not reverse — the stated split. The rival reading this chain does settle is the naive intuition that "the gap is mostly recoverable with better engineering": C1′s GT-15 derivation and the floor argument in this chain's second hop rule that out directly, independent of exactly how the ≈11 points are apportioned among (a)–(c).

Conclusion C5: Pursuing either lever (near-term HX/turbine/cooling refinement at fixed T_hot, or raising T_hot via supercritical steam and eventually chloride salts) changes which actors move and what becomes locked in, on a time horizon of years to a decade or more, in ways that can contradict the efficiency-only framing of C1–C4

C4 (recoverable ≈2–4 pts near-term / locked-in ≈7–9 pts at fixed T_hot; topology ≈+6–7 pts separately) + GT-9 (LCOE falls ~8% at 650 °C) + GT-10? (Gen3 chloride-salt/sCO_2 pathway) + GT-11? (nitrate-salt thermal-stability ceiling) → Actor lens: EPC contractors and turbine OEMs capture the near-term HX/turbine/cooling slice (C4′s ≈2–4 points) through incremental equipment bids with no new materials qualification, while DOE-

funded national labs and a narrower set of specialty-salt and sCO_2-turbine suppliers (GT-10?) are the actors positioned to capture the topology/T_hot lever, because it requires qualifying new corrosion-resistant materials (GT-11?) before any plant operator can bid it into a project → Time lens: the near-term slice is capturable within a normal EPC contract cycle (single-digit years), while the topology/T_hot lever is gated by materials-qualification and first-of-a-kind demonstration risk that the Gen3 program's own structure (GT-10?) implies is still pre-commercial, so on a decade horizon the two levers are not substitutes — a plant ordered today captures only the first → A plant operator who captures only the near-term slice (C4) while a competing dispatchable-power technology (grid-scale batteries paired with PV) continues falling in cost independent of any steam-cycle thermodynamics is a second-order effect this chain does not have ground truths to quantify, but it is a channel by which the entire efficiency gap computed in C1–C4 could become economically moot before the topology lever is commercially qualified *[Assumes: A-13]* — this does not contradict any ground truth in this analysis, so it extends the chain rather than routing back to Phase 2, but it is named here because it bears on how much engineering effort the locked-in and recoverable figures above are actually worth pursuing → None of these downstream effects contradicts GT-1 through GT-15: they bear on the economic and organizational consequences of the efficiency findings, not on the findings themselves, so no routing back to Phase 2 is triggered

Confidence: LOW — carries C4's LOW rating forward and adds two further `?`-marked inputs (GT-10?, GT-11?), both reported-by-delegate. The verification that would remove GT-10?/GT-11? as a cause of the downgrade is opening the NREL Gen3 Liquid Pathway report and the cited chloride-salt corrosion paper directly, neither of which this analysis read at source. This chain is explicitly second-order and scoped as such in the Conclusion; it is not load-bearing for C1–C4's numeric findings, which stand independently of whether this chain's organizational predictions hold.

5. Abandoned Reasoning

Dead End: The 290 °C cold salt tank as the cold reservoir

What was tried: Computing η_Carnot = 1 − T_cold/T_hot using the two salt-tank temperatures directly (565 °C hot tank, 290 °C cold tank) as the reservoir pair, since both are well-sourced, round, and salient numbers (GT-4).

Why abandoned: The cold tank is the return leg of the heat *source* loop, not a heat sink — nothing in the plant rejects waste heat at 290 °C. The arithmetic is well-formed (1 − 563/838 = 32.8%) but the result is not a ceiling on the actual machine: GT-6's real, operating design practice (43.0% gross) already exceeds it. A bound real equipment exceeds has the wrong reservoir pair, not a measurement error — this is the decisive test, and this framing fails it.

What it ruled out: Any later re-derivation that reaches for the tank pair because both numbers are salt-side and convenient. The correct cold reservoir is always the point where the *cycle* rejects heat (chain C1), never a temperature internal to the heat-delivery loop, however tempting its availability.

Dead End: Curzon-Ahlborn as the "more realistic" law-permitted ceiling

What was tried: Using η_CA = 1 − √(T_cold/T_hot) as the ceiling instead of (or alongside) Carnot, on the reasoning offered in the question itself — that it "accounts for the unavoidable finite-rate heat transfer needed to deliver nonzero power" and so is a tighter, more physically honest bound than the classical, zero-power-limit Carnot figure.

Why abandoned: Curzon-Ahlborn is the efficiency of one specific, named engine model — internally reversible, finite-rate coupled to its reservoirs, operated at the power level that *maximizes power output* — not a bound on heat engines generally. GT-6's real design-point efficiency (43.0% wet / 41.2% dry) already exceeds the CA value computed for the same reservoir pair (38.6%/37.3%, chain C1). A plant engineered for efficiency, not power density, is not described by the CA model, and the fact that real equipment already beats it is decisive, not merely suggestive.

What it ruled out: The intuitive but incorrect move of treating "accounts for finite-rate heat transfer" as the same thing as "is a valid ceiling." Finite-rate heat transfer is real and does destroy exergy (this is exactly what GT-15′s exergy-correct ceiling and chain C3′s category (a) and (b) brackets price in), but Curzon-Ahlborn prices it in for the wrong optimization target.

Dead End: An exergy ceiling that bakes in current HX approach-temperature gaps and steam-metallurgy limits

What was tried: Following the question's third suggested candidate — constructing a "law-permitted ceiling" that already incorporates the ≈28 °C salt-to-steam approach gap current heat-exchanger design accepts, and the live-steam-temperature cap current turbine metallurgy imposes, on the reasoning that these constraints are "unavoidable" in practice.

Why abandoned: Approach-temperature gaps and metallurgy limits are engineering and economic conventions (current constraints, per the Assumptions Table's taxonomy), not laws — GT-6 itself documents three different metallurgy/pressure classes (subcritical, supercritical, ultrasupercritical) at three different approach gaps, all operating today or in near-term design studies, which is direct evidence these limits move with investment and materials choice. Folding a movable convention into "the ceiling" produces a number that looks rigorous but is actually just a restatement of current best design practice — the same failure mode as citing best-in-class incumbent practice as a theoretical limit, just with the incumbent figure computed rather than observed.

What it ruled out: Treating "the ceiling" and "the best achievable design today" as the same quantity. They are not: the first is what chain C1 computes (GT-15′s exergy-correct bound), the second is GT-6′s design-point figure, and the distance between them is exactly chain C2′s gap — collapsing them would have made the central question of this analysis unanswerable by definition.

Dead End: Reporting the naive single-reservoir Carnot gap (≈19 points) as the headline figure

What was tried: Early in Phase 4, computing the gap as naive-Carnot-ceiling minus current-practice (62.3% − 43.0% = 19.3 points), matching the treatment a prior theoretical-limit pass on this same domain used.

Why abandoned: Once GT-15′s exergy-correct derivation (chain C1) established that the salt's own temperature glide lowers the correct ceiling by roughly 8 points, continuing to report the naive 19-point figure as "the gap" would have attributed real, structural exergy content of the heat source — exergy that no engine, however perfect, could ever have extracted, because it was never thermodynamically available in the first place — to the plant's own shortfall. That materially changes the headline relative-gap figure (≈21% vs. a prior ≈31%) and therefore the answer to Part 5′s recoverable/locked-in split.

What it ruled out: Any synthesis in §6 that quotes "~19 points" or "~31% relative" as the gap. Both numbers are real and reported (chain C2′s fifth hop, for direct comparability with other treatments of this domain), but they are explicitly labeled as the looser, naive-ceiling figures, not the headline result.

Dead End: Resolving Crescent Dunes' and Gemasolar's exact cycle parameters from secondary sources

What was tried: Multiple WebSearch and WebFetch attempts to establish Crescent Dunes' and Gemasolar's measured or design-point gross/net cycle efficiencies, including direct fetch attempts against NREL's SolarPACES project database and NREL's SAM case-study documentation.

Why abandoned: Both direct-fetch attempts against nrel.gov subdomains (solarpaces.nrel.gov, sam.nrel.gov) failed with DNS resolution errors, and the secondary sources that were reachable disagree with each other: Wikipedia's Crescent Dunes figure ("almost 600 °C" hot tank) is 30–35 °C above the 565 °C-class design-basis literature this analysis otherwise relies on (GT-4–GT-6), and no source reachable by this analysis — primary or secondary — states a measured or design gross/net cycle efficiency for either plant.

What it ruled out: Presenting any specific percentage for Crescent Dunes or Gemasolar in §6. The honest-depth path is naming this as a data gap (chain C2′s final hop, GT-12?/GT-14?), not estimating past it with a number dressed as a fact.

6. Conclusion

Recommended approach: Cite the exergy-correct, non-isothermal-source Carnot ceiling —

≈54.3% wet-cooled / ≈52.3% dry-cooled — as the law-permitted ceiling, not the naive single-reservoir Carnot figure (≈62.3%/60.6%, valid but loose) and not Curzon-Ahlborn (≈38.6%/ 37.3%, not a valid ceiling at all — real design practice already exceeds it) (chain C1). Against that ceiling, current commercial design practice (43.0% wet / 41.2% dry gross; an estimated 35.8–38.3% net after parasitics) sits roughly **11 points below on a gross basis (≈21% relative) and 14–16 points below on a net basis (≈27–31% relative)** (chain C2) — a materially smaller gap than the ≈19–25 points the naive Carnot figure would report, because the naive figure overstates the ceiling by treating the salt as isothermal rather than as the cooling, sensible-heat stream it actually is (chain C2). Of the ≈11-point gross gap, roughly **2–4 points (a quarter to a third) are recoverable with known near-term engineering** at the current 565 °C salt temperature (supercritical steam pressure, tighter HX approach, wet cooling where water is available, better storage dispatch), while roughly **7–9 points (two-thirds to three-quarters) are fundamentally locked in** by the Second Law applied to a real, finite-power heat exchanger and real turbomachinery (chain C4). Raising the salt temperature (toward 650 °C nitrate-salt ultrasupercritical steam, or beyond via chloride-salt/sCO_2 designs) is a separate, substantial lever on *absolute* efficiency, not a way to close this fixed-T_hot gap — and on the one case this analysis could compute, it narrows the *relative* gap too (≈12% at 650 °C against ≈21% at 565 °C) (chain C4).

Key insight: The question "which ceiling is defensible" is not a preliminary to this analysis — it is most of the analysis. Molten salt is not an isothermal reservoir; it is a finite stream that cools 275 °C while delivering heat, and a Carnot ceiling computed at its peak temperature silently assumes away roughly 8 points of "headroom" that was never thermodynamically available to begin with (chain C1). Correcting for this shrinks the apparent gap by nearly half in relative terms (≈21% vs. a naive ≈31%) and, more importantly, inverts the naive intuition that "raising T_hot closes the Carnot gap": raising T_hot resets the whole bracket upward and, if anything, makes the cycle *more* reversible relative to its own new ceiling, but it does not touch the locked-in component of the gap at the original operating point at all (chain C4).

Trade-offs acknowledged: This synthesis accepts a wide, explicitly stated uncertainty band on the gap decomposition (chain C3, LOW confidence) in exchange for not inventing false precision: no plant-specific exergy/heat-and-mass-balance model for a molten-salt tower was located, so categories (a), (b), (c)'s residual, and the cycling portion of (e) are Fermi brackets, not measurements (chain C3). It also accepts a genuine, named data gap on Crescent Dunes' and Gemasolar's actual cycle parameters rather than filling it with numbers dressed as facts (chain C2). Finally, it deliberately leaves the economic question of whether the locked-in and recoverable efficiency figures are worth pursuing at all, given competing dispatchable-generation cost trajectories, as a named but unquantified second-order channel (chain C5) rather than folding an unsupported claim into the headline numbers. It also scopes itself explicitly to steady-state thermodynamic loss mechanisms: reliability and availability losses (unplanned outages, equipment failures, the kind of extended downtime Crescent Dunes itself experienced) are a real, additional gap between nameplate design efficiency and actual annual energy capture that none of categories (a)-(e) (chain C3) quantifies, and this analysis does not estimate it — no chain — flagged assumption only.

Confidence: LOW — the headline recoverable/locked-in split (chain C4) is capped by chain C3's LOW rating, which is itself capped by the Fermi-bracket apportionment (A-9, A-10) rather than a plant-specific exergy model. This is not a weak conclusion in the sense of being poorly reasoned — chain C1, which establishes *which* ceiling is defensible, is rated HIGH, and chain C2, which establishes *how large* the gap is, is rated MEDIUM; both rest on read-at-source primary documents (Sandia SAND2011-9320, SAND2002-0120, the DOE SunShot fact sheet) — it is LOW specifically on the *apportionment* of the gap across mechanisms, which is the one piece of Part 4 that would need a per-component exergy-destruction study or a measured commercial-scale operating record to tighten, exactly as chain C3's own confidence line states. The direction of every finding here (ceiling choice, gap size, dominance of the locked-in component, T_hot as a separate lever) is robust to that remaining uncertainty; only the precise point-estimates within the 2–4 / 7–9 split would move.